



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Freelancer Project Hub

The domain of the Project:

MERN Stack Web Application Development

Team Mentors (and their designation):

PRASANNA KUNDU - Software Engineer at Numenor

Team Members:

1. Ms. Gaddam Sujala B.Tech , 3rd year pursuing

Period of the project

August 2025 to September 25



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Declaration

This project, "Freelancer Project Hub," was developed independently by me, Sujala Gaddam, with guidance from mentors. The project's purpose was to gain practical, hands-on experience by building a full-stack, industry-relevant application.

I declare that the work presented here is my own and has significantly improved my skills in software development.

Team Members:

1. Ms. Gaddam Sujala Signature
G.Sujala

Mentor's Name Prasanna Kundu

Designation— Software Engineer at numenor

Prof. Radhakumari

Executive Director & Founder

SURE Trust



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Executive Summary

- This report outlines the development and architecture of the **Freelancer Project Hub**, a web application designed to create a streamlined, centralized workspace for clients and freelancers.
- The platform addresses the common challenges of project management by providing a dedicated environment for collaboration, task management, and secure transactions.
- The application's core features include robust user authentication with role-based access, real-time messaging, a Kanban-style task board, cloud-based file storage, and a secure payment flow.
- The system is built on a **MERN stack** (MongoDB, Express, React, Node.js) and is designed with a decoupled client-server architecture, ensuring scalability and maintainability.



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Introduction

- The Freelancer Project Hub aims to simplify the post-hiring workflow between clients and freelancers.
- Unlike large marketplaces, this platform focuses on the active project phase, providing a clean, distraction-free environment for collaboration.
- The project's scope encompasses user authentication, project creation, a proposal system, and a comprehensive project workspace.
- The application's core innovation lies in its integration of a real-time communication system with a structured task and deliverable management system.



Project Objectives

- ☑ To develop a full-stack, Single-Page Application (SPA) using the MERN stack.
- ☑ To implement a secure, JSON Web Token (JWT)-based user authentication system with separate client and freelancer roles.
- ☑ To create a real-time, project-specific messaging system using **Socket.IO**.
- ☑ To integrate a cloud-based file storage solution, **Cloudinary**, for managing user profile pictures and project deliverables.
- ☑ To build a comprehensive task management feature where users can create, update, and delete tasks with a 'to-do', 'in-progress', and 'done' status.
- ☑ To design a user-friendly and responsive front end using React with **React Router DOM** for navigation.



Methodology and Results

- **Methods/Technology used:**

The project follows a decoupled architecture where the frontend (React) and backend (Node.js/Express) operate as separate services. The backend serves as a REST API and handles all data logic and interactions with a **MongoDB Atlas** database. Real-time communication is managed by a separate Socket.IO server. User-uploaded files are handled by **Multer** and stored on Cloudinary.

- **Tools/Software used:**

- **Version Control:** Git & GitHub
- **Code Editor:** Visual Studio Code
- **Backend:** Node.js, Express.js, Nodemon (for development)
- **Frontend:** React (with Vite), React Router DOM, Axios
- **Database:** MongoDB Atlas (cloud database)
- **Real-time Communication:** Socket.IO
- **Authentication:** JSON Web Tokens (JWT) and Bcryptjs
- **Cloud Storage:** Cloudinary (for file and image uploads)
- **API Testing:** Postman (for testing API endpoints)

- **Data collection approach (if applicable):** The data for this application is not collected from an external source but is generated directly by the users of the platform. The application is a live system where data is created, read, updated, and deleted based on user actions. For example, when a client posts a project or a freelancer sends a proposal, this information is collected via the application's forms and stored in the **MongoDB Atlas** database.

- **Project Architecture (Clearly describe the architecture):**

- **Final project working screenshots along with supporting explanation :**



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The screenshot shows the FreelancerHub dashboard for a user named 'sujji'. The dashboard is divided into several sections:

- FreelancerHub** (Header)
- Welcome, sujji!** (Greeting)
- Find new projects and manage your work.** (Sub-header)
- Search projects...** (Search bar)
- Dashboard** (Left sidebar menu)
- Profile Section:** Displays the user's profile picture, name 'sujji', role 'Web Developer', a star rating of 3.7/5, and a list of skills: web development, arts, 45 wpm, DSA, ml. It also shows the user's earnings of \$1,810 and an 'Edit Profile' button.
- Your Stats** (Right sidebar): A table showing key metrics.

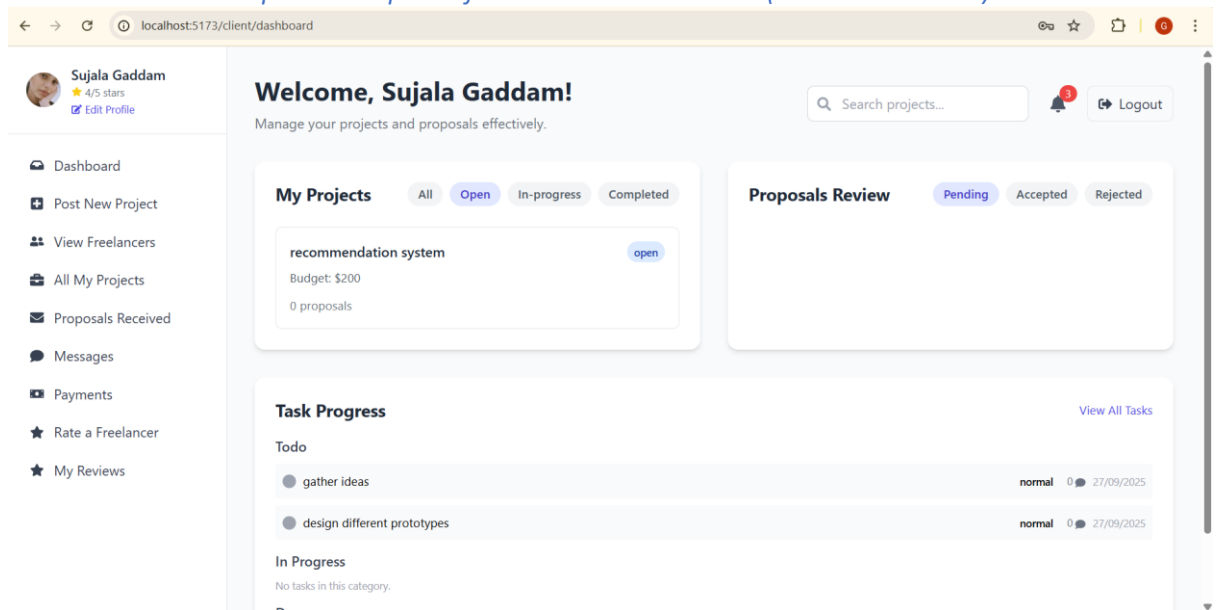
Your Stats	
Proposals Sent	2
Active Projects	2
Earnings	\$1,810
Client Rating	3.7/5
- Job Recommendations** (Main content area): A list of open projects. The first project is 'recommendation system' with a budget of \$200. It includes the client's name 'Sujala Gaddam', a rating of 4/5, and a description: 'i need a recommendation system for my shopping app'. There is a 'View Client Details' link and an 'Apply Now' button.
- My Active Projects** (Right sidebar): A list of projects where the user has been hired. The first project is 'website design' with a budget of \$200. It includes a description: 'i need an website to store and manage ...' and a 'Collaborate' button.

This is the central hub for the freelancer. It provides a quick overview of their professional activity and helps them find new work.

- **User Profile and Stats:** The top section displays the freelancer's profile picture, name, role, and a star rating. On the right, a "Your Stats" card shows key metrics like the number of proposals sent, active projects, and total earnings, giving them a clear view of their business performance.
- **Job Recommendations:** The main area is dedicated to a curated list of open projects from clients. This section is designed to help the freelancer easily find and apply for new opportunities. Each project card shows a title, a brief description, budget, and the client's name and rating.
- **Active Projects:** A separate section highlights projects where the freelancer has been hired. This provides a direct link to the collaboration space for ongoing work, streamlining their workflow.



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This dashboard is designed to give the client full control over their projects and the hiring process.

- **Profile and Quick Actions:** The top left of the dashboard shows the client's profile and average rating. The main header provides a search bar to filter projects and a notification bell to alert them of new proposals.
- **Project Management:** The "My Projects" section is the core of this dashboard. It lists all projects posted by the client, with options to filter them by status (e.g., "open," "in-progress," "completed"). This allows the client to quickly check the status of all their work.
- **Proposals Review:** The "Proposals Review" section provides a dedicated area for the client to manage incoming applications for their projects. They can review and accept or reject proposals directly from the dashboard.
- **Task Progress:** The "Task Progress" section gives the client a quick, at-a-glance view of the status of tasks across all of their active projects, keeping them informed about the work being done.

- **Project GitHub Link:**

FRONTEND: <https://github.com/sujji-123/FreelancerProjectHub-frontend>

BACKEND: <https://github.com/sujji-123/FreelancerProjectHub-backend>



Learning and Reflection

Developing the Freelancer Project Hub was a comprehensive and highly valuable learning experience. The primary challenge was the seamless integration of numerous disparate systems into a single, cohesive platform. I learned how to:

- **Implement Real-Time Functionality:** Integrating **Socket.IO** to create the real-time chat and task updates was a major achievement. This taught me about WebSockets and event-driven architecture.
- **Manage Cloud-Based Services:** I gained practical experience in working with cloud-based services like **MongoDB Atlas** for database management and **Cloudinary** for scalable file storage.
- **Strengthen Security Skills:** Implementing **JWT-based authentication** and using **bcryptjs** for password hashing provided a deep understanding of secure user management.
- **Enhance Frontend Development:** I developed a robust frontend using **React** and **React Router DOM**, which taught me how to create a dynamic, multi-page user experience within a Single-Page Application (SPA) structure.



Conclusion and Future Scope

- **CONCLUSION:**

The Freelancer Project Hub project successfully achieved its core objectives, resulting in a fully functional and robust freelance collaboration platform. By utilizing the MERN stack, the project demonstrated a strong grasp of modern web development principles, including decoupled architecture and real-time communication. The integration of various technologies, such as Socket.IO for live chat, Cloudinary for file management, and JWT for secure authentication, was a significant learning experience that solidified my practical skills in building a cohesive, end-to-end application. Ultimately, this project serves as a practical showcase of my ability to design, develop, and deploy a complex, industry-relevant application, preparing me for future challenges in software development.

- **Future scope of this project:**

- **Enhanced Payment System:** I will integrate a more robust payment gateway like Stripe, complete with webhooks for transaction confirmation, as mentioned in the original project notes.

- **Advanced User Profiles:** Implementation of a public user profile page that displays a user's projects, ratings, and reviews.

- **Email Notifications:** Adding a feature to send email notifications for important events such as new messages or project updates.

- **Mobile Application:** Developing a native mobile app using a framework like React Native to provide users with on-the-go access.



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